



gamania | **Vyin AI**

Gamania: Advancing AI capabilities for enterprises with Google Cloud

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GOOGLE CLOUD RESULTS

- ✓ Enhances data analytics efficiency by 1,600x with [BigQuery](#).
- ✓ Enables real-time data analytics for more timely business decision-making
- ✓ Supports fast, highly reliable AI services through [Google Kubernetes Engine](#)
- ✓ Reduces AI model prototype development time from three months to two days

By using Google Cloud to deploy its centralized data warehouse and run analytics efficiently, Gamania is able to improve its subsidiaries' business performance and swiftly develop innovative AI solutions leveraging Vertex AI.

[Gamania](#) has been a leading figure in Taiwan's online PC gaming industry for more than 30 years. Offering some of the most popular games such as [MapleStory](#) and [Lineage](#), Gamania quickly became a listed company on Taiwan's stock market and has built a comprehensive online ecosystem by expanding its business into various sectors, including payment, ecommerce, media, digital business, and innovation.

In 2020, Gamania set a new mission of becoming an international digital corporation with industry-leading technologies. First, the company implemented data-driven decision-making across its subsidiaries. As its analytics capabilities and the amount of its data grew, Gamania gradually established a solid foundation for artificial intelligence (AI) innovation and launched more than a dozen of AI solutions internally. Following these successful internal releases, Gamania founded Vyin AI in 2024, a brand that provides enterprise-grade AI services.



“We believe that big data and AI are becoming a fundamental part in every business’ daily operations,” explains Benjamin Chen, Chief Strategy Officer and Head of AI Innovation Lab at Gamania. “That’s why we’ve been consolidating our data and AI capabilities by founding an AI innovation lab and recruiting top talents. Now, we’re ready to share our AI expertise with companies across different industries and help them optimize their operational performance.”

When Gamania wanted to roll out data-driven strategies in the corporate group, its more than 10 subsidiaries were using relational databases deployed in various environments to store or process data. Transferring data between each subsidiary was complicated, and there was a lack of parallel computing architectures for efficient big data analytics. As a result, Gamania needed to build a centralized data warehouse in a short period of time, so that it could quickly embark on its data-driven transformation journey.

After reviewing different solutions, the corporate group chose to employ [Google Cloud](#) because of its better performance in processing large amounts of data, as well as its ESG commitment that's in line with Gamania's core values.

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Enhancing data analytics efficiency by 1,600x with BigQuery

In 2020, Gamania's big data team laid out a unified architecture of data collection using [Dataflow](#) and [Dataproc](#) for all Gamania's subsidiaries and began to store all its business data in [Cloud Storage](#). Based on the data, the team quickly built a centralized data lake in [BigQuery](#) to enable speedy data retrieval and big data analytics. Chasel Su, Head of Big Data Center and Co-Head of AI Innovation Lab at Gamania, notes that with key technologies like column-oriented storage, serverless autoscaling, distributed nodes, query optimization, BigQuery not only supports 1.8x faster query

speed than other similar offerings, but also allows the quick setup of Gamania's centralized data warehouse through seamless integration with datasets in Cloud Storage and data sources like [Google Ads](#) and [Google Analytics](#) that Gamania's subsidiaries rely heavily on.

Prior to the adoption of Google Cloud, whenever Gamania needed to conduct cross-subsidiary data analytics, its data scientists needed more than one week to gather and process data from databases deployed in various environments. Now, even Gamania's non-engineer employees can use BigQuery to run the analytics they need on their own and retrieve reports within a few minutes.

This way, Gamania is able to execute effective and timely data-driven decision-making. For example, when its operations team runs campaigns, it needs to adjust strategies based on consumers' engagement. Before using BigQuery, the team had to wait for two days to receive data for campaign performance evaluation. The bigger a campaign was, the longer the team had to wait for the data.

Now, by leveraging Dataflow to stream data in real time to BigQuery for analytics, the Gamania team can monitor how consumers respond to their campaigns instantly and swiftly change their strategies when needed.

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Su. “We’re now able to process 100x more data with 1,600x higher analytics efficiency, which translates into better business decision-making.”



Supporting highly reliable AI solutions with abundant computing resources

Having consolidated its centralized data storage and analytics, Gamania decided to maximize the value of its data through the machine learning (ML) technology. For its ecommerce and online media business, it developed several AI recommendation systems by leveraging Dataproc to process data and tag products automatically, and GPU resources in [Compute Engine](#) to train ML models. To provide the most relevant content, Gamania also employs [Pub/Sub](#) and Dataflow to support real-time data streaming and ensure that its recommendation system can reflect users’ latest behavior within three seconds.

Since its recommendation system is trained with a large number of data and capable of generating recommended content based on user behavior under different circumstances like seasons and time, Gamania is able to provide hyperpersonalized experiences and has improved the conversion rate on its ecommerce sites by up to 17x, as well as the interaction rate on social media by up to 300 percent.

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Head of Big Data Center and Co-Head of AI Innovation Lab,
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Su points out that the abundant computing resources in Google Cloud play an important role in the development and deployment of Gamania's AI recommendation system. As Compute Engine provides many ready-to-use GPU drivers or toolkits, it's rather effortless for the Gamania team to set up ML training environments. Furthermore, the team deploys the recommendation system's API in [Google Kubernetes Engine](#) (GKE), which offers great scalability that guarantees response time of less than 500 milliseconds even during high traffic periods such as sales seasons.

"For AI solutions that require a lot of computing power to process and transfer data like our recommendation system, it is extremely crucial to have powerful computing resources," he adds. "Compute Engine and GKE have facilitated our development process and ensured excellent service reliability."



Enabling fast AI service development with Vertex AI

Besides the AI recommendation system, Gamania has developed more than a dozen AI solutions deployed in GKE for internal use, including a customer service chatbot named Vyin Brain that it launched in two of its most popular games and reached a resolution rate of 97 percent.

One of Vyin Brain's major features is the ability to detect user context, which is supported by a number of complex analyses like intention, emotion, and entity relationship analysis. To facilitate the development process, Gamania needed to first merge all the analysis models and resolve the integration issues. To that end, the team used the pretrained models and templates in [Vertex AI](#) to quickly build a prototype and identify the potential problems, which made it much easier to optimize and finetune its self-developed models afterwards.

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concept in a short period of time and avoid many trials and errors.”

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“The ready-to-use AI models in Vertex AI are very helpful for us to develop our own AI solutions, because we can build a proof of concept in a short period of time and avoid many trials and errors,” notes Su. “For example, we leveraged the text-to-speech model in Vertex AI to develop a voice imitation prototype model within two days, instead of three months if we had trained the model from scratch.”

On top of that, the generative AI capabilities of Vertex AI has helped Gamania enhance work efficiency. Its data analysts now use [Gemini 1.5 Flash](#) in Vertex AI to batch process semantic analytics in BigQuery, which has greatly shortened their time to generate analytics reports without asking AI engineers to build a semantic analytics model.



Introducing AI to more sectors for greater benefits

With its strong expertise in AI innovation, Gamania will start providing AI solutions to enterprises in various sectors across Taiwan, Southeast Asia, and North America from the fourth quarter of 2024 through its newly established brand Vyin AI. At the same time, the corporate group plans to expand its use of the Gemini models to further increase its employees’ productivity, such as image generation and workflow optimization.

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develop innovative AI solutions. We’re confident that our enterprise customers will enjoy the same or even greater benefits from our AI services deployed in Google Cloud.”

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Gamania is a multidisciplinary company that has expanded its business into games, payment, ecommerce, media, digital business, and innovation. Having its business development centered around AI, big data, and digital platforms, Gamania has built a comprehensive online ecosystem. In 2024, Gamania’s AI innovation lab launched Vyin AI, a brand that provides diverse AI solutions to enterprises.

Industry: Games

Location: Taiwan

Products: [BigQuery](#), [Vertex AI](#), [Cloud Storage](#), [Compute Engine](#), [Dataflow](#), [Dataproc](#), [Google Kubernetes Engine](#), [Pub/Sub](#)

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